# Day 31 Task – Access Control Fundamentals

**Access control is an important concept in cybersecurity that determines who can access computer systems, networks, applications, and sensitive information. It ensures that only authorized users and devices can access specific resources according to predefined security policies.**

**Access control helps organizations prevent unauthorized access, protect confidential information, and maintain accountability. One of the most widely used security frameworks for managing access is the AAA model, which stands for Authentication, Authorization, and Accounting.**

**These three components work together to verify user identities, determine access permissions, and record user activities.**

**Objective**

**The objective of this task is to understand the fundamentals of access control and examine the three components of the AAA security model: Authentication, Authorization, and Accounting.**

**It also aims to explain their individual roles, differences, and importance in protecting organizational systems and resources.**

**What Is Access Control?**

**Access control is a security mechanism that regulates which users or systems are allowed to access particular resources and what actions they are permitted to perform.**

**For example, in an organization, an employee may be allowed to view and edit project documents, while only an administrator may modify system configurations or manage user accounts.**

**Access control helps enforce security policies and ensures that users receive only the permissions required to perform their responsibilities.**

**Understanding the AAA Security Model**

**AAA stands for Authentication, Authorization, and Accounting. It is a security framework used to manage and monitor access to systems and network resources.**

**Authentication**

**Authentication is the process of verifying the identity of a user, device, or system before granting access.**

**It answers the question: Who are you?**

**Authentication can be performed using different methods:**

- **Something you know: Passwords, PINs, or security answers.**
- **Something you have: Security tokens, smart cards, or authentication applications.**

- **Something you are: Biometric characteristics such as fingerprints or facial recognition.**

**Multifactor authentication combines two or more different authentication factors to provide stronger protection.**

**Example: When an employee logs into a company account using a password and a one-time verification code, the system verifies the employee's identity through two authentication factors.**

**Authorization**

**Authorization determines what an authenticated user is allowed to access or perform within a system.**

**It answers the question: What are you allowed to do?**

**After authentication, the system checks the user's assigned permissions and security policies before allowing access to specific resources.**

**For example:**

- **An ordinary employee may view and edit assigned documents.**
- **A manager may access departmental reports.**
- **An administrator may create accounts and modify system settings.**

**Authorization helps prevent users from accessing information or performing actions beyond their assigned responsibilities.**

**Accounting**

**Accounting involves recording and monitoring user activities and resource usage within a system or network.**

**It answers the question: What did you do?**

**Accounting records may include:**

- **Login and logout times.**
- **Accessed resources.**
- **Commands executed.**
- **Network usage.**
- **Session duration.**
- **Changes made to system configurations.**

**These records help administrators monitor activities, investigate security incidents, identify unusual behavior, and maintain accountability.**

**Example: When an employee accesses a company database, the system may record the username, access time, resource accessed, and actions performed.**

**Difference Between Authentication, Authorization, and Accounting**

| Feature | Authentication | Authorization | Accounting |
|---|---|---|---|
| Meaning | Verifies identity | Determines access permissions | Records user activities |
| Main question | Who are you? | What can you access? | What did you do? |
| Purpose | Identity verification | Access management | Activity monitoring |
| When it occurs | Before authorization | After authentication | During and after resource usage |
| Example | Entering a password | Accessing an approved file | Recording file access |
| Security benefit | Prevents unauthorized logins | Prevents unauthorized actions | Supports auditing and accountability |

**How the AAA Process Works**

**The AAA process generally follows these steps:**

**Authentication: A user attempts to access a system by providing credentials or other identity verification information. The system verifies the user's identity.**

**Authorization: Once the user's identity is verified, the system checks the permissions assigned to that account and determines which resources can be accessed.**

**Accounting: The system records relevant activities, such as login times, resource usage, and actions performed during the session.**

**This process helps organizations manage access securely while maintaining records for monitoring and auditing.**

**Role-Based Access Control (RBAC)**

**Role-Based Access Control is an access management approach in which permissions are assigned according to a user's organizational role rather than individually assigning every permission.**

**For example, an organization may define the following roles:**

- **Employee: Can access assigned work documents and applications.**
- **Manager: Can access departmental information and approve certain requests.**
- **Administrator: Can manage users, permissions, and system configurations.**

**RBAC simplifies permission management and supports the principle of least privilege.**

**Common Authentication Methods**

**Password-Based Authentication: Users provide a username and password to verify their identity.**

**Biometric Authentication: Uses physical or behavioral characteristics, such as fingerprints or facial recognition.**

**Token-Based Authentication: Uses a physical or digital token to verify identity.**

**Multifactor Authentication (MFA): Combines multiple authentication factors to strengthen account security.**

**Certificate-Based Authentication: Uses digital certificates to verify the identity of users or devices.**

**AAA in Network Security**

**AAA is commonly used in network environments to control access to routers, switches, wireless networks, and remote access services.**

**Two protocols frequently associated with AAA are:**

**RADIUS (Remote Authentication Dial-In User Service): A protocol commonly used to provide centralized authentication, authorization, and accounting for network access.**

**TACACS+ (Terminal Access Controller Access-Control System Plus): A protocol commonly used for centralized access control and administration of network devices. It separates authentication, authorization, and accounting functions.**

**These protocols help organizations manage user access to network infrastructure through centralized security systems.**

**Importance of the AAA Model**

**The AAA model provides several security benefits:**

- **Verifies user identities before granting access.**
- **Ensures users receive only authorized permissions.**
- **Records user activities for monitoring and auditing.**
- **Helps detect unauthorized access attempts.**
- **Supports investigation of suspicious activities.**
- **Improves accountability within an organization.**
- **Strengthens access management across networks and systems.**

**Security Best Practices**

**Organizations should implement strong authentication methods, including multifactor authentication for important accounts.**

**Access permissions should follow the principle of least privilege, ensuring that users receive only the access necessary for their responsibilities.**

**Accounting records should be maintained securely and reviewed regularly to identify unusual login patterns, unauthorized access, and suspicious activities.**

**Administrative privileges should be restricted to authorized personnel. Organizations should also review user accounts periodically and remove unnecessary access when employees change roles or leave the organization.**

**Access control is a fundamental part of cybersecurity that protects systems and information from unauthorized access. The AAA model provides a structured approach to access management through Authentication, Authorization, and Accounting.**

**Authentication verifies identities, authorization determines permissions, and accounting records user activities. By implementing strong authentication, appropriate access permissions, activity monitoring, and regular security reviews, organizations can strengthen their access control systems and protect sensitive information.**